

Analog (LP) input via a Creative SB1240 X-Fi HD

Analog (LP) input via a Creative SB1240 X-Fi HD

1. What works with no changes
2. Three settings that must be changed by hand
 - 2.1 Sample width — `omdrc_cdin_bits`
 - 2.2 Sample rate
 - 2.3 Recording source — `omdrc_audio_capture_recsrc`
3. The behaviour that does not carry over: the silence gate
4. Stopping it / handing the chain back
 - 4.1 Release the output, keep the daemon running
 - 4.2 Stop the daemon
 - 4.3 Turn it off across reboots
5. Suggested first run
6. Settings summary
7. Files referenced

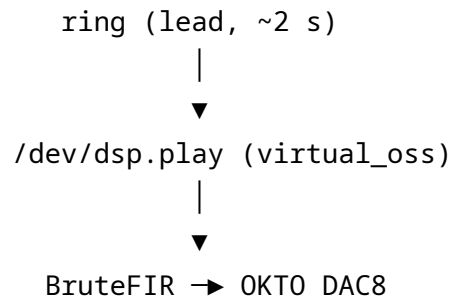
Analog (LP) input via a Creative SB1240 X-Fi HD

Purpose: run vinyl through the DRC chain for a listening test, reusing `omdrc-cdin` — the bridge written for the ESI U24 XL S/PDIF interface.

Short answer: `cdin` is not ESI-specific and will work, but selecting the card in the web configuration page is *not* sufficient. Three settings must be changed by hand, and one behaviour (the silence gate) does not apply to an analog source at all — so you must know how to hand the chain back manually.

Signal path for the test:

LP → Lehmann phono stage → SB1240 Line in → `/dev/dsp.capture`
| `omdrc-cdin`
▼



Use the **Line** input, not Phono: the Lehmann already applies RIAA and gain, and the card's phono path would apply both a second time.

1. What works with no changes

`cdin/src/main.c` opens whatever `--in` points at. The ESI appears only in comments and defaults — there is no device-specific code.

The web **Configuration** page writes the card selection and reconciles the role symlinks, so `/dev/dsp.capture` follows the SB1240 across replugs:

Step	File
page + endpoint	<code>omdrc-ctrl/src/app.py</code>
writes <code>omdrc_audio_capture=</code> , runs <code>omdrc_audio reconcile</code>	<code>scripts/omdrc-config-helper.py</code> (<code>freebsd_apply</code> , ~line 84)
resolves the card, creates <code>/dev/dsp.capture</code>	<code>etc/rc.d/omdrc_audio.in</code>
reads <code>/dev/dsp.capture</code> by default (<code>omdrc_cdin_in</code>)	<code>cdin/rc.d/omdrc_cdin.in</code>

That part is genuinely one click.

2. Three settings that must be changed by hand

None of these are exposed in the web UI; all are `rc.conf` variables consumed by `cdin/rc.d/omdrc_cdin.in` and `etc/rc.d/omdrc_audio.in`.

2.1 Sample width — `omdrc_cdin_bits`

Default is **24**, and that is an ESI fact, not a neutral default: the U24 XL's capture endpoint is 24-bit S-LE and nothing else. `omdrc_audio` sets `bitperfect=1` on the capture card (`omdrc_audio_capture_sysctls`), so there is

no format feeder to hide a mismatch behind — a wrong width fails the open outright.

Find what the SB1240 actually exposes, then set it:

```
cat /dev/sndstat
sysctl dev.pcm.<N>           # <N> = the capture card's pcm unit
sysrc omdrc_cdin_bits=24     # or 16, per the above
```

2.2 Sample rate

The rc script does **not** pass `--rate`, so `cdin` uses its own default of **44100**. FreeBSD's `uaudio` fixes one (rate, bits) pair at attach time, so the card must be attached at the rate you ask `cdin` for. To run the test at another rate, pass it through the escape hatch:

```
sysrc omdrc_cdin_flags="--rate 48000"
```

and line up the driver side (`hw.usb.uaudio.default_rate / default_bits`, or the per-unit sysctls) to match.

2.3 Recording source — `omdrc_audio_capture_recsrc`

This one silently captures the wrong jack. `audio_apply_recsrc()` in `etc/rc.d/omdrc_audio.in` (~line 450) treats `auto` as “prefer `pcm2 dig1 dig2 dig3`” — i.e. the *digital* input, which is correct for the ESI. The SB1240 has an S/PDIF input too, so `auto` will happily select it and you will record silence while the LP plays.

Pin the analog input explicitly. Find its real mixer name first — only sources tagged `rec` are selectable:

```
mixer -f /dev/mixer.capture    # look for the line tagged "rec"
sysrc omdrc_audio_capture_recsrc=line
service omdrc_audio reconcile
```

3. The behaviour that does not carry over: the silence gate

`cdin/src/gate.h` releases the output device when the wire has carried **exact zeros** for `omdrc_cdin_idle_after` ms — “a CD’s silence is literally `0x0000`, so there is no threshold to tune on the sample side”.

An analog ADC's noise floor is never zero. Therefore:

- GATE_IDLE is **never** reached;
- omdrc-cdin holds /dev/dsp.play **indefinitely**;
- MPD and mpv cannot get the chain back until you intervene.

The carrier detector (cdin/src/carrier.h) does not rescue this either. It tests whether frames are arriving at rate; a free-running ADC is always clocked, so carrier_live() is permanently true. That verdict is *correct* — it is simply measuring something that only varies for a clock-slaved S/PDIF receiver.

Consequence: treat cdin-on-analog as something you start for a listening session and hand back afterwards, not something you leave enabled the way the CD input is.

4. Stopping it / handing the chain back

Three levels, least disruptive first.

4.1 Release the output, keep the daemon running

This is the one you normally want — it frees /dev/dsp.play for MPD without restarting anything. It is SIGHUP, wrapped as an rc verb (omdrc_cdin_release in cdin/rc.d/omdrc_cdin.in; it is also what drc.sh does before stopping virtual_oss):

```
service omdrc_cdin release
```

By hand, if you started the daemon yourself rather than via the service:

```
kill -HUP "$(cat "${TMPDIR:-/tmp}/omdrc_cdin-${id -un}.pid")"
```

Pidfile locations (set in cdin/rc.d/omdrc_cdin.in, ~line 124):

Started as	omdrc_cdin_pidfile
root (service omdrc_cdin start)	/var/run/omdrc_cdin/omdrc_cdin.pid
unprivileged (service omdrc_cdin onestart)	\${TMPDIR:-/tmp}/omdrc_cdin-\${id -un}.pid

4.2 Stop the daemon

```
service omdrc_cdin stop
```

or onestop, if you used onestart

A foreground run (see §5) stops on Ctrl-C: SIGINT/SIGTERM set `cdin_stop`, which interrupts the blocking transfers rather than letting them retry (`cdin/src/cdin.h`).

4.3 Turn it off across reboots

```
sysrc omdrc_cdin_enable=NO
```

Do not reach for stopping `virtual_oss` as a way of freeing the chain: on this box that intermittently wedges it unkillable and pins `cuse.ko`, needing a reboot. Release `cdin` instead.

5. Suggested first run

Do **not** enable the service for the first attempt. Run it in the foreground so a failed format negotiation is loud rather than buried in a log:

```
omdrc-cdin --in /dev/dsp.capture --out /dev/dsp.play \  
           --rate 44100 --bits <what sndstat reports> \  
           --lead 2000 -d -s 10 -l /tmp/cdin-lp.log
```

Checks while a side plays:

1. **starves must stay 0.** Each one is an audible dropout. The SB1240 is a different USB topology from the ESI and may have a different stall floor, so do not assume the ESI's calibrated `--lead` transfers — recalibrate per the procedure in `cdin/README.md`.
2. **The drift field** settles after ~5 minutes and reports measured ppm. Two independent crystals (SB1240 ADC vs OKTO) is the same situation as the CD case, and the `lead ÷ drift` arithmetic is unchanged: ~11 h at 50 ppm against a 2000 ms lead. An LP side is ~25 minutes, so drift cannot cause a discontinuity within a side. There is no resampler in the path.
3. **The DAC must be powered on.** The OKTO enumerates and streams with its front panel off — a run with the panel dark measures a sleeping converter and proves nothing.

Note the ~2 s lead is also the transport lag: dropping the needle is heard about two seconds later, and lifting it runs on for about two seconds.

6. Settings summary

```
# card selection – or use the web Configuration page
sysrc omdrc_audio_capture="<SB1240 as it appears in /dev/sndstat>"
sysrc omdrc_audio_capture_recsrc=line          # NOT auto: auto picks
                                                S/PDIF

# cdin
sysrc omdrc_cdin_bits=<16|24>                  # must match the endpoint
                                                exactly
sysrc omdrc_cdin_flags="--rate 44100"         # only if not 44100
sysrc omdrc_cdin_lead=2000                    # recalibrate for this
                                                card

service omdrc_audio reconcile
service omdrc_cdin start
# ... listen ...
service omdrc_cdin release                    # give /dev/dsp.play back
                                                to MPD
service omdrc_cdin stop
```

7. Files referenced

Path	Why it matters here
cdin/rc.d/omdrc_cdin.in	every omdrc_cdin_* variable; the release verb; pidfile paths
etc/rc.d/omdrc_audio.in	card→role resolution, /dev/dsp.capture, audio_apply_recsrc(), bitperfect=1
cdin/src/gate.h	the exact-zero silence gate that will never fire on analog
cdin/src/carrier.h	the rate-based carrier detector; always live on a free-running ADC
cdin/src/cdin.h	cdin_stop — how SIGINT/SIGTERM interrupt transfers
cdin/src/main.c	option parsing, width negotiation, the [stats] line
cdin/README.md	lead calibration procedure
scripts/omdrc-config-helper.py	what the web Configuration page actually writes

Path	Why it matters here
omdrc-ctrl/src/app.py	the CD input card / log tailing in the panel